Multi-scale Hybrid Mamba Network for Lightweight Semantic Segmentation

Haisheng Wang[a], Yingying Zhao[a], Jiu-Cheng Xie[a], Guangwei Gao[b]

[a] *College of Automation, Nanjing University of Posts and Telecommunications, Nanjing, China*
[b] *School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China*

Abstract

Real-time semantic segmentation is a core component of environment perception in autonomous driving systems. Models based on convolutional neural networks (CNNs) and Transformers have achieved significant progress in this task. However, both paradigms struggle to efficiently capture multi-scale features under strict lightweight computational constraints. To address these limitations, this paper enhances a state space model with linear computational complexity and integrates it with the Multi-branch Feature Aggregation Block (MFAB), proposing a Multi-scale Hybrid Mamba Network (MHMNet). The proposed hybrid architecture adopts an asymmetric encoder–decoder design. In the encoder, MFAB enables cross-scale and multi-dimensional feature extraction, while the decoder employs a Hybrid Mamba module to collaboratively model local details and global contextual information. Subsequently, the Gated Feature Selection Head (GFSH) is introduced to suppress noisy channel interference and enhance the contribution of discriminative features, producing the final segmentation output. This design significantly reduces computational complexity while preserving segmentation accuracy. The proposed MHMNet contains only 0.77M parameters and achieves mean Intersection-over-Union (mIoU) scores of 71.48% and 69.5% on the Cityscapes and CamVid benchmarks, respectively. Experimental results demonstrate that the proposed model maintains competitive

Email address: whs400@163.com (Haisheng Wang)

segmentation accuracy under a substantially reduced parameter budget, effectively alleviating the trade-off between accuracy and efficiency in lightweight segmentation models and achieving a favorable balance between performance and computational complexity.

Keywords: Real-time semantic segmentation, Lightweight network, Mamba, Multi-scale Feature

1. Introduction

Semantic segmentation serves as a fundamental task for environment perception and scene understanding in autonomous driving systems. Its core objective is to assign a precise and consistent semantic label to each pixel in an input image, thereby providing fine-grained perceptual cues for downstream modules such as path planning, behavior prediction, and safety-critical decision-making in complex urban scenarios. Since autonomous driving systems are typically deployed on resource-constrained embedded platforms, stringent requirements are imposed on real-time inference capability, operational stability, and computational efficiency. Consequently, achieving high-accuracy semantic segmentation under limited computational budgets has become a critical and challenging research problem in this field.

The rapid advancement of deep learning has facilitated the widespread adoption of CNN-based methods in real-time semantic segmentation scenarios. Benefiting from the local receptive fields and inherent spatial inductive biases of convolutional operators, such approaches exhibit notable advantages in computational efficiency and model robustness. For instance, BiSeNetV2 [1] introduces a dual-branch architecture that decouples high-level semantic understanding from low-level boundary detail modeling, significantly improving inference speed through lightweight module design. DDRNet [2] proposes a dual-resolution parallel architecture, where a high-resolution branch preserves fine-grained spatial structural information while a low-resolution branch performs efficient semantic modeling, thereby enhancing the representation of road structures and ob-

ject contours without compromising real-time performance. Nevertheless, these CNN-based methods are inherently constrained by the fixed local receptive fields of convolutional operators, which limit their ability to model long-range dependencies and cross-region semantic consistency. As a result, they are prone to semantic fragmentation and insufficient global coherence in large-scale urban scenes.

To compensate for the limited global modeling capability of CNNs, Transformer [3] architectures based on self-attention mechanisms have been progressively introduced into the semantic segmentation domain, providing a new paradigm for long-range dependency modeling. SegFormer [4] adopts a hierarchical Transformer encoder that captures long-range dependencies through multi-stage feature abstraction, achieving a favorable balance between accuracy and efficiency with a lightweight decoder design. SegNext [5] further addresses the high computational cost of Transformers in real-time scenarios by simplifying and restructuring multi-scale context aggregation, substantially reducing model complexity while retaining global modeling capability. However, the self-attention mechanism intrinsically exhibits quadratic computational complexity, which leads to considerable computational and memory overhead when processing high-resolution inputs, thereby limiting the practical deployment of Transformer-based models in real-world autonomous driving systems.

Recently, the Mamba State Space Models (SSMs) have emerged as a promising alternative for efficient long-range dependency modeling. By leveraging a selective scan mechanism, Mamba achieves effective modeling of long-sequence dependencies while maintaining linear computational complexity, striking a balance between expressive power and inference efficiency. Compared with Transformers, Mamba demonstrates superior computational scalability in high-resolution feature modeling, offering a new research direction for lightweight and real-time urban scene semantic segmentation. However, directly applying state space models to visual semantic segmentation remains challenging due to insufficient multi-scale feature representation and limited local spatial structure modeling capability, necessitating effective integration with the spatial inductive

priors of convolutional networks.

Lightweight semantic segmentation in complex urban scenes faces a fundamental tension under strict parameter and real-time constraints. Accurate delineation of boundaries, thin structures, and small objects demands strong local spatial priors and multi-scale representations, whereas globally consistent prediction requires effective long-range dependency modeling. However, purely strengthening local multi-branch designs easily exhausts limited model capacity, while attention-based global modules often incur prohibitive computational and memory overhead. Therefore, an effective lightweight framework should unify efficient multi-scale local encoding with scalable global context propagation to improve both fine details and holistic semantics. Motivated by the above analysis, this paper proposes a Multi-scale Hybrid Mamba Network(MHMNet), a lightweight semantic segmentation framework that synergistically integrates the local spatial inductive bias of CNNs with the efficient long-range dependency modeling capability of Mamba state space models within a unified architecture. The main contributions of this work are summarized as follows:

- We propose a multi-scale hybrid feature encoding mechanism by introducing a Multi-branch Feature Aggregation Block (MFAB) into the encoder. MFAB facilitates efficient local feature extraction and cross-scale fusion through multi-path interactions, enabling more effective utilization of spatial information across different receptive fields in complex urban scenes.

- We design a Hybrid Mamba decoding module that integrates state space modeling with local feature processing mechanisms. This module enables collaborative modeling of local details and global contextual semantics, strengthening global semantic consistency while maintaining linear computational complexity.

- A Gated Feature Selection Head (GFSH) is constructed by introducing a channel-wise gating mechanism to adaptively select decoded features. This module effectively suppresses noise-induced channel interference and

emphasizes the contribution of discriminative features to final segmentation predictions, further improving segmentation accuracy and runtime stability in lightweight settings.

2. Related work

2.1. CNN and Attention mechanism

Convolutional neural networks have long served as a fundamental backbone for semantic segmentation, owing to their strong capability in local feature extraction, spatial structure modeling, and efficient inference. Recent research has primarily focused on enhancing representation capacity and improving real-time performance. In the high-quality segmentation regime, SFNet [6] introduces learnable multi-scale feature encoding and structured context modeling to strengthen spatial detail representation and semantic consistency, demonstrating robust performance in complex urban scenes. In the lightweight setting, multi-branch architectures remain a dominant design paradigm. PPLiteSeg [7] reduces computational overhead through network structure redesign and efficient channel interaction mechanisms, while PIDNet [8] adopts a parallel semantic–detail modeling strategy to achieve a favorable balance between inference speed and segmentation accuracy. Despite their effectiveness in local feature modeling and real-time deployment, these CNN-based approaches are inherently constrained by fixed receptive fields, which limit their ability to capture long-range dependencies and maintain global semantic consistency.

To address the limited global modeling capability of CNNs, attention mechanisms have been widely explored in semantic segmentation and other vision tasks. Several studies aim to reduce the computational complexity of attention, with some achieving linear complexity through structural approximations or reformulations. However, many of these approaches rely on frequent feature shifting or reshaping operations, which introduce additional memory access overhead and latency during inference, thereby reducing their suitability for real-time applications. MSCA [9] enhances spatial context modeling via large-

kernel and multi-scale attention designs and achieves promising segmentation performance, but its large kernel operations and multi-scale attention structure incur increased inference cost and are less hardware-friendly.

Further efforts have focused on simplifying attention mechanisms to improve practical efficiency. External Attention [10] adopts a compact formulation by introducing external learnable parameters to replace conventional key–value computations, enabling attention modeling with only a small number of linear transformations. Building upon this idea, GPU-Friendly Attention (GFA) [11] removes the head-splitting operation in multi-head attention and replaces it with group-based normalization, making the attention computation more amenable to GPU parallelization. Although these methods improve computational efficiency and deployment friendliness, they still rely on explicit attention computation and feature reweighting, which pose challenges under high-resolution inputs and strict real-time constraints. These limitations motivate the exploration of more efficient global modeling paradigms for lightweight semantic segmentation.

2.2. State Space Models for Semantic Segmentation

State space models originated from the control theory domain, where the core idea is to describe the evolution of dynamic systems via linear state update equations. They have been widely applied in sequence modeling, signal processing, and related tasks. With the development of structured state space approaches, the Structured State Space Sequence Model (S4)systematically introduced this framework into deep learning. By discretizing continuous-time state equations and employing structured parameterization, S4 enables effective modeling of long-range dependencies while maintaining linear computational complexity, laying a crucial theoretical and methodological foundation for modern SSMs research. Compared to traditional recurrent neural networks and self-attention mechanisms, S4 exhibits superior stability and computational scalability in long-sequence modeling tasks.

Building upon S4, Mamba further proposed a data-driven state selection mechanism, allowing the state space model to dynamically adjust the state

update path according to input features. This design significantly enhances the model's feature representation capacity and modeling flexibility, overcoming the inherent limitations of traditional SSMs in complex feature modeling, while still maintaining linear computational complexity and learning more discriminative representations. Inspired by this approach, Vision Mamba and VMamba [12, 13] extended one-dimensional state space modeling to two-dimensional image structures by introducing cross-scan and multi-directional scan strategies. This adaptation allows SSMs to better capture complex spatial dependencies in visual tasks, and experiments on multiple visual benchmarks demonstrate performance improvements over conventional convolutional networks and self-attention-based models.

These advances have further motivated the exploration of SSMs in semantic segmentation. Several studies have integrated Mamba modules into segmentation network architectures to enhance global context modeling. For instance, VM-UNet [14] incorporates visual state space modules into the classical encoder-decoder framework to establish long-range dependencies across regions effectively, while RS3Mamba [15] employs a dual-branch network design to collaboratively model local detail features and global semantic information, achieving significant improvements in remote sensing image segmentation. Moreover, MAFMamba [16] introduces an adaptive multi-scale fusion design with global–local enhancement and cross-attention-based feature fusion for high-resolution remote-sensing segmentation. In contrast, our method targets real-time urban-scene segmentation under much stricter lightweight constraints, and differs mechanistically by emphasizing a compact CNN–SSM hybrid local–global collaboration with linear-time context modeling instead of attention-centric cross-level adaptive fusion. These works demonstrate that state space models can provide strong global semantic modeling capability while remaining computationally efficient. However, many SSM-based segmentation methods still depend on comparatively heavy backbones or attention-centric fusion pipelines, and lightweight real-time urban-scene segmentation with jointly enhanced multi-scale representation and long-range context modeling remains

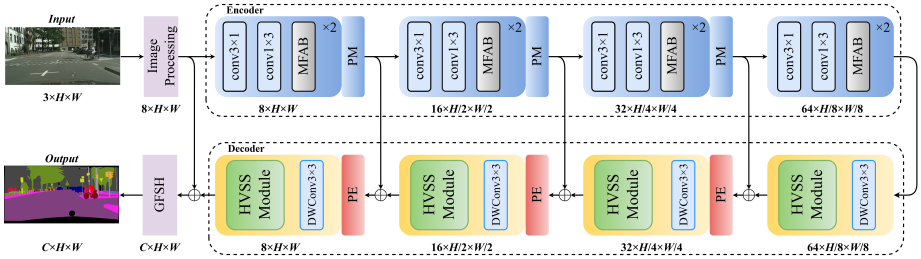

Figure 1: The complete architecture of the proposed Multi-scale Hybrid Mamba Network.

insufficiently studied. Systematic architectural designs and mechanism-level explorations for collaboratively modeling global dependencies and local detail features within a unified network remain limited, leaving important research space for further improving the overall performance of lightweight semantic segmentation models.

3. PROPOSED METHOD

3.1. Overview Network Architecture

As illustrated in Figure 1, MHMNet follows the UNet [17] design paradigm with an asymmetric architecture, consisting of an image preprocessing module, a backbone encoder, a decoder based on Hybrid Visual State Space Module (HVSS), and a Gated Feature Selection Head (GFSH).

Given an input image $X \in \mathbb{R}^{3\times H\times W}$, the model first applies a preprocessing convolutional module to expand the channel dimension to 8, constructing an initial feature representation, resulting in a feature map $X' \in \mathbb{R}^{8\times H\times W}$. The encoder is composed of four sequentially stacked encoding blocks. Each block contains two consecutive convolutional units, where each unit includes a 3×1 convolution, a 1×3 convolution, and a Multi-branch Feature Aggregation Block (MFAB). This component leverages multi-kernel convolutions and attention mechanisms to decouple spatial feature extraction from channel transformations, thereby capturing cross-level, multi-dimensional detail features with

a lightweight parameter footprint while maintaining a sufficient receptive field. Patch Merging (PM) operations are applied after the first three encoding blocks, halving the spatial resolution and doubling the channel dimension to aggregate features via downsampling.

The decoder mirrors the encoder with four stages. Each decoder block consists of a 3×3 depthwise separable convolution followed by an HVSS module. The depthwise separable convolution performs lightweight feature transformation, while the HVSS module jointly models global and local context by combining local convolutional priors with state space modeling. To compensate for spatial detail loss during downsampling, skip connections are employed to directly fuse features from the corresponding encoder stage via element-wise addition. At the beginning of the last three decoding stages, Patch Expanding (PE) operations are applied to progressively restore spatial resolution and reconstruct high-resolution feature maps.

Finally, the decoder output is processed by the Gated Feature Selection Segmentation Head, which maps the feature channels to the target number of classes C while further refining segmentation performance, producing the final prediction map $P \in \mathbb{R}^{C \times H \times W}$.

3.2. Multi-branch Feature Aggregation Block (MFAB)

Existing semantic segmentation methods have widely incorporated multi-scale feature extraction at the encoder stage to enhance contextual representation. However, most of them primarily focus on spatial scale aggregation across different receptive fields, while largely overlooking explicit interaction modeling between channel-wise semantics and spatial dimensions. As a result, multi-scale features are often insufficiently decoupled and collaboratively exploited. In complex urban scenes, objects at different scales are usually accompanied by significant structural variations and semantic hierarchy changes, where simple scale stacking or sequential fusion tends to introduce feature redundancy or suppress discriminative cues. Motivated by these observations, we propose a Multi-branch Feature Aggregation Block (MFAB), which introduces multi-

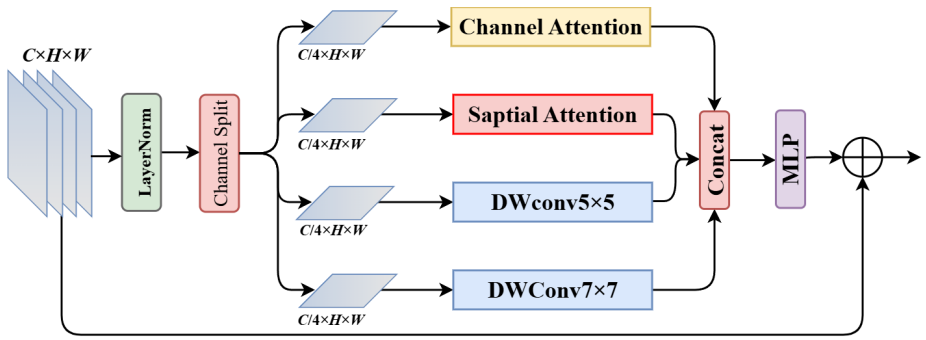

Figure 2: The architecture of the proposed Multi-branch Feature Aggregation Block (MFAB).

path feature modeling and cross-dimensional interactions at the encoder stage to jointly encode multi-scale spatial features and channel-wise semantic information, thereby enhancing the discriminability and robustness of the learned representations.

MFAB is shown in Figure 2. Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, the module first applies Layer Normalization to stabilize feature distributions, and then evenly splits the normalized feature along the channel dimension into four sub-features:

$$[X_1, X_2, X_3, X_4] = \mathrm{Split}(\mathrm{LN}(X)), \tag{1}$$

where each sub-feature $X_i \in \mathbb{R}^{\frac{C}{4}\times H\times W}$ is responsible for modeling complementary feature characteristics. Specifically, the first two branches introduce attention-based feature modulation to model complementary dependencies. The first branch employs a channel attention mechanism to capture global channel-wise relationships and adaptively recalibrate the importance of different channels, while the second branch incorporates a spatial attention mechanism to explicitly model spatial correlations and emphasize informative regions. The corresponding operations can be formulated as:

$$\widetilde{X}_1 = \mathcal{A}_{\mathrm{ch}}(X_1), \quad \widetilde{X}_2 = \mathcal{A}_{\mathrm{sp}}(X_2), \tag{2}$$

where $\mathcal{A}_{\mathrm{ch}}(\cdot)$ and $\mathcal{A}_{\mathrm{sp}}(\cdot)$ denote the channel attention and spatial attention op-

erators, respectively, and $\widetilde{X}_1$ and $\widetilde{X}_2$ represent the enhanced feature maps after attention-based reweighting. The third and fourth branches adopt depthwise convolutions with kernel sizes of 3×3 and 5×5, respectively, to capture local contextual information under different receptive fields with low computational overhead:

$$\widetilde{X}_3 = \mathrm{DWConv}_{3\times3}(X_3), \quad \widetilde{X}_4 = \mathrm{DWConv}_{5\times5}(X_4). \tag{3}$$

The enhanced features from all branches are concatenated along the channel dimension to form a unified representation

$$F = \mathrm{Concat}(\widetilde{X}_1, \widetilde{X}_2, \widetilde{X}_3, \widetilde{X}_4), \tag{4}$$

which is further transformed by a lightweight MLP to enable cross-channel interaction and nonlinear feature reorganization. Finally, a residual connection is applied to obtain the module output:

$$Y = X + \mathrm{MLP}(F). \tag{5}$$

From a mechanistic perspective, MFAB enforces complementary feature specialization by splitting the input into parallel sub-branches. Channel attention performs global semantic recalibration, suppressing redundant channels while amplifying category-discriminative responses; spatial attention provides location-sensitive modulation to emphasize informative regions and sharpen boundary-related cues. In parallel, multi-kernel depthwise convolutions aggregate local context with different receptive fields, capturing scale-dependent structures at low cost. Together, these three components act synergistically to form a compact yet expressive representation.

3.3. Hybrid Visual State Space Module (HVSS)

Current lightweight semantic segmentation methods still exhibit clear limitations in urban scenes. Convolution-based models suffer from the slow growth of the effective receptive field, making it difficult to establish sufficient long-range dependencies in low-contrast regions or areas with ambiguous semantic

boundaries, which in turn hampers holistic scene understanding. Attention-based models, while capable of capturing global context, incur computational costs that grow substantially with image resolution, rendering it challenging to balance accuracy and efficiency in real-time applications. Moreover, some linear-complexity global modeling approaches, such as direct serialization-based operations, may weaken the intrinsic 2D spatial structure when applied to images, thereby degrading the discrimination of small objects and fine-grained details.

Motivated by these issues, we build upon the efficient global modeling paradigm of Mamba and further propose a Hybrid Visual State Space (HVSS) module to construct more robust and discriminative global semantic representations within a lightweight network framework. Unlike state-space designs that rely solely on single-sequence modeling or simple directional scanning, HVSS organically integrates local convolutional modeling with two-dimensional state space modeling. This design preserves linear-time complexity and hardware-friendly computation, while effectively enhancing cross-region semantic interaction.

An overview of the HVSS module is provided in Fig. 3. At each decoder stage, a feature map generated by a 3×3 depthwise separable convolution, denoted as $X \in \mathbb{R}^{C \times H \times W}$, is first normalized by LayerNorm to stabilize the feature distribution. It is then passed through a linear projection to expand the channel dimension to twice its original size, increasing the representational capacity. The expanded features are further processed by a 3×3 convolution followed by a SiLU activation to strengthen local spatial cues, and are subsequently fed into the Hybrid 2D Selective Scan (HSS2D) module for joint global-context and local-structure modeling, yielding an enhanced feature:

$$X_1 = \mathrm{HSS2D}(\mathrm{SiLU}(\mathrm{Conv}_{3\times3}(\mathrm{Linear}(X)))) \,. \tag{6}$$

Here, $\mathrm{SiLU}(\cdot)$ denotes the SiLU activation, $\mathrm{Conv}_{n\times n}(\cdot)$ denotes an $n \times n$ convolution, and $\mathrm{Linear}(\cdot)$ represents a linear transformation.

In parallel, to preserve the original information of the input, we introduce an auxiliary branch that applies a linear projection followed by an activation to

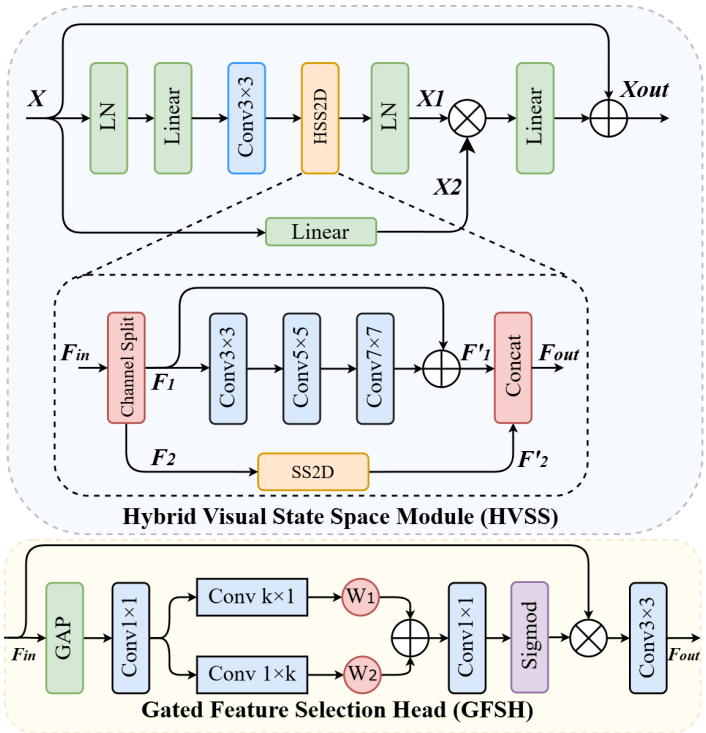

Figure 3: The architecture of the Hybrid Visual State Space Module (HVSS) and the Gated Feature Selection Head Module (GFSH).

produce a complementary feature:

$$X_2 = \mathrm{SiLU}(\mathrm{Linear}(X))\,. \tag{7}$$

The two branches are fused via element-wise multiplication, and the fused representation is projected back to the original channel dimension and added to the input through a residual connection:

$$X_{\mathrm{out}} = X + \mathrm{Linear}(X_1 \otimes X_2)\,, \tag{8}$$

where $\otimes$ denotes element-wise multiplication.

This design preserves local spatial structures while leveraging HSS2D to capture global dependencies, achieving global–local feature enhancement. It improves discriminability under complex backgrounds and maintains computa-

tional efficiency, providing effective representations for low-contrast regions in urban road-scene segmentation.

HSS2D serves as the core component of the HVSS module and follows a hybrid modeling paradigm that organically integrates the local spatial modeling capability of convolutions with the long-range dependency modeling strength of state space models. Specifically, the input feature F_{in} is first partitioned along the channel dimension via a channel split into two subspaces. The first is a local enhancement branch F_1, which sequentially applies multi-scale convolutions with kernel sizes 3×3, 5×5, and 7×7 to progressively aggregate local textures and boundary cues, and then adopts a residual connection to form a stable locally enhanced representation F_1'. The second is a global modeling branch F_2, which is fed into SS2D to capture long-range dependencies and global context with linear complexity, producing F_2'. Finally, the two branch features are integrated by channel concatenation to yield the output F_{out}, enabling complementary fusion of local details and global semantics within a unified framework and mitigating the representational limitations incurred by relying on either convolution-only or sequence-only modeling.This process can be formulated as follows:

$$F_1,\, F_2 = \mathrm{Split}(F_{in}), \tag{9}$$

$$\tilde{F}_1 = \mathrm{Conv}_{7\times7}\Big(\mathrm{Conv}_{5\times5}\big(\mathrm{Conv}_{3\times3}(F_1)\big)\Big), \tag{10}$$

$$F_1' = F_1 + \tilde{F}_1, \tag{11}$$

$$F_2' = \mathrm{SS2D}(F_2), \tag{12}$$

$$F_{out} = \mathrm{Concat}\big(F_1',\, F_2'\big). \tag{13}$$

SS2D is a fundamental state-space unit for global dependency modeling. It aims to transform a 2D feature map into multi-directional sequences, perform global interactions in the sequence domain via a state space model, and then remap the processed sequences back to the original spatial layout. Specifically, given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, SS2D first applies scan expanding

by traversing the feature map in four directions—left-to-right, right-to-left, top-to-bottom, and bottom-to-top—to obtain a sequence representation $x \in \mathbb{R}^{C \times L}$, where $L = H \times W$ denotes the sequence length. This operation enables the network to capture latent global semantic relations embedded in the spatial structure from different directional perspectives.

Given the resulting sequence, SS2D employs the S6 state space module for efficient long-range modeling. In this process, three independent linear projections are applied to the input sequence to produce the discretization step size Δ, the input projection coefficient B, and the output projection coefficient C:

$$\Delta,\, B,\, C = \mathrm{Linear}_i(x), \quad i = 1, 2, 3. \tag{14}$$

Following the discretization of the state space model, the state transition matrix $\overline{A}$ and the discretized input mapping matrix $\overline{B}$ are computed as

$$\overline{A} = \exp(\Delta A), \tag{15}$$

$$\overline{B} = (\Delta A)^{-1}\big(\exp(\Delta A) - I\big) \cdot \Delta B, \tag{16}$$

where $\exp(\cdot)$ denotes the matrix exponential and I is the identity matrix. The latent state h_t progressively accumulates semantic dependencies over time, and is updated recursively by

$$h_t = \overline{A} h_{t-1} + \overline{B} x_t, \tag{17}$$

with the output computed as

$$y_t = C h_t + D x_t, \tag{18}$$

where D is a learnable input-residual term that preserves local information while constructing global dependencies. Concatenating the outputs over all time steps yields the complete sequence output:

$$y = [y_1, y_2, \ldots, y_t, \ldots, y_L]. \tag{19}$$

Finally, the scan merging procedure remaps the four directional sequence outputs back to a 2D feature map, followed by directional alignment and aggregation to obtain the final output $Y \in \mathbb{R}^{C \times H \times W}$. By combining scan expanding, S6-based sequence modeling, and scan merging, SS2D establishes global

contextual interactions with linear complexity while maintaining an effective representation of the underlying spatial structure.

In our implementation, the above SS2D recursion is realized through a Mamba-style selective scan kernel. Specifically, the state updates and output accumulations are computed by an optimized scan operator with input-dependent coefficients, rather than by explicitly forming transition matrices or evaluating matrix exponential terms. This realization preserves the intended discrete state space dynamics, while providing a numerically stable and hardware-efficient execution that scales linearly with the sequence length.

3.4. Gated Feature Selection Head (GFSH)

In semantic segmentation, high-level feature maps produced by the backbone are rich in semantic cues; however, their channel-wise contributions to the final prediction are inherently unequal. Some channels may encode highly discriminative patterns that are strongly correlated with specific categories, whereas others may predominantly capture redundant background responses or task-irrelevant noise. Directly using all channels for prediction can therefore hinder the model capacity due to interference from non-discriminative features, especially when dealing with objects exhibiting ambiguous boundaries or large appearance variations.

To address this issue and generate accurate pixel-level segmentation maps, we design a Gated Feature Selection Head, as illustrated in Fig. 3. The key idea is to leverage global contextual information to calibrate the input features and employ a lightweight sub-network to dynamically emphasize informative channels while suppressing redundant or noisy ones. This gating mechanism enables the network to adaptively assess channel importance and re-scale feature responses dynamically, thereby improving segmentation accuracy as well as boundary fidelity.

Given an input feature map $F_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, the proposed GFSH module constructs a content-adaptive gating signal to reweight feature responses before prediction. As shown in Fig. 3, we first summarize global context via global

average pooling and project it with a 1×1 convolution:

$$u = \mathrm{Conv}_{1\times 1}\big(\mathrm{GAP}(F_{\mathrm{in}})\big). \tag{20}$$

To model complementary dependencies with negligible overhead, u is processed by two lightweight asymmetric convolution branches:

$$a = \mathrm{Conv}_{k\times 1}(u), \qquad b = \mathrm{Conv}_{1\times k}(u). \tag{21}$$

The two responses are then adaptively combined using learnable weights W_1 and W_2:

$$s = W_1 \odot a + W_2 \odot b, \tag{22}$$

where $\odot$ denotes element-wise multiplication. A subsequent 1×1 convolution generates the gating map, followed by a sigmoid activation to constrain the gate into $(0, 1)$:

$$G = \sigma(\mathrm{Conv}_{1\times 1}(s))\,. \tag{23}$$

Finally, the gate is applied to the input feature through channel-wise modulation, and a 3×3 convolution is used to produce the output for segmentation prediction:

$$F_{\mathrm{out}} = \mathrm{Conv}_{3\times 3}\big(F \odot G\big). \tag{24}$$

By conditioning G on the globally pooled representation, GFSH enables instance-aware reweighting of channel responses, while the asymmetric branches provide an efficient mechanism to capture complementary patterns for gate generation. The learnable combination (W_1, W_2) further improves flexibility, allowing the head to adaptively balance the two branches during training.

4. Experiments

In this section, we conduct extensive experiments on two widely used street-scene benchmark datasets to rigorously evaluate the performance of the proposed MHMNet. We first introduce the experimental setup in detail, including

Table 1: The details of the model settings.

Dataset	Cityscapes	CamVid
Batch Size	6	8
Main Loss	CrossEntropy	OHEM CrossEntropy
Auxiliary Loss	Focal Loss	
Optimization method	SGD (momentum 0.9)	Adam (momentum 0.9)
Weight decay	1×10^{-4}	2×10^{-4}
Initial learning rate	4.5×10^{-2}	1×10^{-3}
Learning rate policy	Poly	

dataset descriptions, loss functions, implementation details, and hardware configurations. Subsequently, a series of ablation studies is carried out to systematically analyze the impact of each key component within MHMNet. Finally, we compare MHMNet with several state-of-the-art semantic segmentation methods through both quantitative metrics and qualitative visualizations, demonstrating its clear advantages in segmentation accuracy, computational efficiency, and inference speed.

4.1. Datasets

Cityscapes [18]: Cityscapes is a standard benchmark for semantic understanding in urban driving scenes. It provides 5,000 finely annotated images collected across 50 cities, with a native resolution of 2048×1024. Following the official protocol, 2,975 images are used for training, 500 for validation, and 1,525 for testing. While the dataset defines 34 semantic classes, evaluation is typically reported on 19 classes using mean Intersection over Union (mIoU). Since the test split contains only images without public labels, performance on the test set is obtained by submitting predictions to the official evaluation server.

CamVid [19]: CamVid is an early but widely adopted urban scene segmentation dataset captured from an on-board vehicle viewpoint. It includes

701 images of size 960×720 with pixel-level annotations and is commonly partitioned into 367 training, 101 validation, and 233 test images. In line with prior work, results are usually reported over 11 semantic categories, covering the most frequent objects and scene elements in road environments.

4.2. Implementation Details

All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 2080Ti GPU. The dataset-specific configurations and training hyperparameters are summarized in Table 1. For CamVid, images are resized to 360×480 and trained with 11 classes using a batch size of 8 and an initial learning rate of $lr_{\text{initial}} = 1 \times 10^{-3}$. For Cityscapes, the input resolution is set to 512×1024 with 19 classes, using a batch size of 6 and $lr_{\text{initial}} = 4.5 \times 10^{-2}$. We train both datasets for 1000 epochs and adopt a polynomial learning-rate schedule:

$$lr = lr_{\text{initial}} \left(1 - \frac{iteration}{max_iteration}\right)^{0.9}, \tag{25}$$

where the learning rate is linearly warmed up during the first 500 iterations and then decayed according to the above rule. Standard data augmentation is applied, including random horizontal flipping and random scaling in the range $[0.5, 2.0]$.

Following Table 1, the main loss for Cityscapes is the standard cross-entropy loss, whereas CamVid employs an OHEM-based cross-entropy to emphasize hard pixels. For both datasets, we additionally apply Focal Loss as an auxiliary supervision term. The overall training objective is defined as

$$loss = loss_{\text{main}} + \lambda\, loss_{\text{aux}}, \tag{26}$$

with $\lambda = 0.1$. We follow a unified training protocol and keep key hyperparameters fixed across all experiments and ablations. The auxiliary-loss weight λ is set to $\lambda = 0.1$ to provide additional supervision without overwhelming the main objective, and the same value is used for both datasets for consistency. For optimization, Cityscapes is trained using SGD with momentum 0.9 and weight decay 1×10^{-4}, while CamVid is trained using Adam with weight decay 2×10^{-4}.

Table 2: Ablation studies of the MFAB module on the CamVid dataset.

Methods	**Param.** (K)↓	**FLOPs** (G)↓	**Speed** (FPS)↑	**mIoU** (%)↑
CA only	698	3.50	315	$67.90^{0.05\uparrow}$
SA only	702	3.51	318	67.85
CNN only	730	3.68	303	$68.20^{0.30\uparrow}$
CA+SA+CNN†	774	3.81	292	$69.50^{1.65\uparrow}$

Table 3: Ablation studies of the HVSS module on the CamVid dataset.

Methods	**Param.** (K)↓	**FLOPs** (G)↓	**Speed** (FPS)↑	**mIoU** (%)↑
CNN	715	3.15	298	68.96
Swin Block	1044	5.07	230	$69.12^{0.16\uparrow}$
VSS	883	4.45	282	$69.23^{0.27\uparrow}$
HVSS†	774	3.81	292	$69.50^{0.54\uparrow}$

Model parameters are updated via backpropagation at each iteration, and the learning rate is adjusted according to the aforementioned schedule.

4.3. Ablation Studies

In this section, we conduct a series of ablation studies to systematically examine the contribution of each component in the proposed MHMNet. All variants are trained under the same protocol on the CamVid training split and evaluated on the corresponding test split, with segmentation metrics reported for comparison.

Ablation Studies of MFAB : Table 2 summarizes ablation results on CamVid to examine the effectiveness of the proposed Multi-branch Feature Aggregation Block (MFAB) for multi-scale hybrid feature encoding. MFAB is designed to enable efficient local feature extraction and cross-scale fusion via multi-path interactions, thereby improving the utilization of spatial cues across

Table 4: Ablation Studies of the GFSH module on the CamVid dataset.

Methods	**Param.** (K)↓	**FLOPs** (G)↓	**Speed** (FPS)↑	**mIoU** (%)↑
–	766	3.75	300	68.93
CNN	770	3.77	290	$69.15^{0.22\uparrow}$
GFSH†	774	3.81	292	$69.50^{0.35\uparrow}$

different receptive fields in complex urban scenes. To validate the necessity of the multi-branch formulation, we compare the full MFAB against several single-path variants that retain only one component, i.e., channel attention (CA only), spatial attention (SA only), or the convolutional branch (CNN only). In particular, our full MFAB configuration (CA+SA+CNN) is highlighted by the red dagger symbol (†) to denote the proposed design.

The results show that all single-path designs yield inferior segmentation accuracy compared with the complete MFAB. Specifically, CA only and SA only obtain 67.90% and 67.85% mIoU, respectively, suggesting that relying on a single attention mechanism is insufficient to capture diverse spatial structures. The CNN-only variant improves performance to 68.20% mIoU by emphasizing local convolutional modeling, yet it still falls short of the full configuration. In contrast, MFAB with CA+SA+CNN (marked with †) achieves the best performance of 69.50% mIoU while remaining lightweight (774K parameters and 3.81G FLOPs) and preserving real-time throughput (292 FPS). These observations verify the complementary nature of multi-path interactions: attention-based cues enhance feature selectivity, whereas the convolutional branch enriches multi-scale local representations, and their integration leads to more discriminative features for urban-scene segmentation.

Ablation Studies of the HVSS Module: Table 3 presents ablation results on CamVid by replacing the proposed HVSS module with different fusion alternatives to verify its effectiveness. Specifically, the first column lists the

module used to substitute HVSS at the same location while keeping the rest of the network unchanged. CNN denotes a conventional convolutional block that models local spatial patterns; Swin Block refers to a lightweight window-based self-attention block adapted from Swin Transformer [20]; and VSS represents a standard state space module for global sequence modeling.

As shown in the table, substituting HVSS with a pure convolutional block yields 68.96% mIoU, indicating that local modeling alone is insufficient for capturing long-range semantics in urban scenes. Introducing attention-based global interactions via the Swin Block improves performance to 69.12% mIoU, yet it incurs a notable increase in complexity (1044K parameters and 5.07G FLOPs) and a substantial drop in throughput (230 FPS). Replacing HVSS with the standard state space design (VSS) further enhances accuracy to 69.23% mIoU, but remains more expensive than HVSS (883K parameters and 4.45G FLOPs) and is still inferior in accuracy. In contrast, the proposed HVSS (marked with the red dagger) achieves the best performance of 69.50% mIoU, while maintaining a favorable efficiency–accuracy balance with only 774K parameters, 3.81G FLOPs, and 292 FPS. These results demonstrate that HVSS effectively integrates state space modeling with local feature processing, enabling collaborative modeling of global contextual semantics and local details, and thus delivers superior segmentation performance under lightweight constraints.

Ablation Studies of the GFSH Module: As presented in Table 4, we conduct an ablation study to evaluate the proposed Gated Feature Selection Head (GFSH). Here, "–" denotes the baseline that uses the original segmentation head without any additional feature selection module. We further include a convolutional refinement variant (CNN) as a stronger baseline, and adopt the proposed GFSH, which introduces a lightweight channel-wise gating mechanism to adaptively reweight decoded features.

Compared with the baseline head, which achieves 68.93% mIoU with 766K parameters, 3.75G FLOPs, and 300 FPS, inserting a CNN refinement block yields a modest gain of +0.22 mIoU, while increasing computation and reducing throughput to 290 FPS. In contrast, GFSH attains the highest accuracy

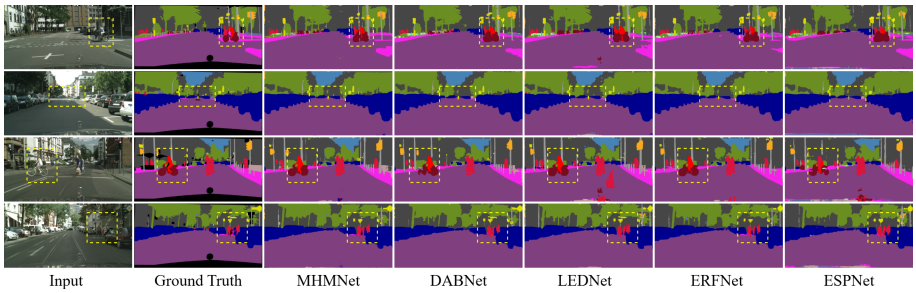

Figure 4: Visual comparisons on the Cityscapes dataset. From left to right are original input images, ground truths, segmentation results from our **MHMNet**, DABNet [50], LEDNet [32], ERFNet [26], and ESPNet [56].

of 69.50% mIoU (+0.35 over the baseline) with only a marginal overhead of 774K parameters and 3.81G FLOPs, while maintaining real-time speed at 292 FPS. These results indicate that channel-wise gating effectively recalibrates decoded representations by attenuating less informative responses and amplifying discriminative channels, thereby improving segmentation accuracy and runtime robustness under lightweight constraints.

The above ablation results indicate that the accuracy improvements are obtained with tightly controlled computational overhead under lightweight budgets. Across MFAB, HVSS, and GFSH, each component yields consistent mIoU gains while keeping the model compact and preserving real-time throughput. Notably, the proposed SSM-based global modeling attains stronger gains than attention-based substitutes in our setting without introducing excessive additional computation, and the lightweight feature selection further improves accuracy with minimal extra overhead. Overall, these findings suggest that the performance gains arise from mechanism-driven improvements in representation efficiency rather than from increasing model complexity.

4.4. Comparisons with Advanced Models

We benchmark our approach against representative state-of-the-art semantic segmentation methods on the Cityscapes and CamVid datasets. The results demonstrate that our model achieves a favorable trade-off between segmentation

Table 5: Comparisons with the state-of-the-arts on the Cityscapes test dataset.

	Methods	Year	Resolution	Backbone	Param. (M)↓	FLOPs (G)↓	GPU	Speed (FPS)↑	mIoU (%)↑
Large Size	DenseASPP [21]	2018	512×512	DenseNet	35.70	632.9	Titan XP	12	80.6
	DANet [22]	2019	1024×1024	ResNet-101	66.60	1298.8	Titan XP	4	81.5
	SETR [23]	2021	768×768	ViT-Large	318.30	-	1080Ti	0.5	82.2
	SegFormer [4]	2021	1024×2048	MiT-B5	122.8	-	-	-	82.4
	HRVIT [24]	2022	1024×1024	HRVIT	28.6	66.8	-	-	83.2
	DDPS-SF [25]	2023	1024×2048	MiT-B5	84.70	1447.6	1080Ti	2.5	84.0
Medium Size	ERFNet [26]	2017	512×1024	No	2.10	-	Titan X	42	68.0
	ESPNet-v2 [27]	2019	512×1024	No	3.49	2.7	Titan Xp	80	66.2
	DFANet [28]	2019	1024×1024	Xeption	7.80	3.4	Titan X	100	71.3
	STDC1-50 [29]	2021	512×1024	-	8.40	-	1080Ti	87	71.9
	BiSeNetV2 [1]	2021	512×1024	Xception	3.40	21.2	1080Ti	92	72.6
	MLFNet [30]	2023	512×1024	ResNet-34	13.03	15.5	Titan XP	72	72.1
	SegTransConv [31]	2024	512×1024	STDC	7.00	10.2	3090	57	72.0
Small Size	LEDNet [32]	2019	512×1024	No	0.94	12.6	1080Ti	78	70.6
	CGNet [33]	2020	360×640	No	0.50	6.0	V100	17.6	64.8
	NDNet [34]	2021	1024×2048	No	0.50	14.0	2080Ti	40	65.3
	SCMNet [35]	2021	512×1024	No	1.20	18.2	2080Ti	92	68.3
	LRDNet [36]	2021	512×1024	No	0.66	16.0	1080Ti	77	70.1
	AGFNet [37]	2022	512×1024	No	1.12	14.4	2080Ti	60	70.1
	PDBNet [38]	2022	512×1024	No	1.82	15.0	2080Ti	50	69.5
	SGCPNet [39]	2022	1024×2048	MobileNet	0.61	4.5	1080Ti	80	70.9
	MSCFNet [40]	2022	512×1024	No	1.15	17.1	Titan XP	50	71.9
	PCNet [41]	2022	1024×2048	Scratch	1.63	11.8	2080Ti	72	72.7
	BANet [42]	2023	512×1024	No	0.72	6.69	3080	83.2	70.1
	SFRSeg [43]	2023	512×1024	No	1.60	19.6	2080Ti	98	70.6
	FBSNet [44]	2023	512×1024	No	0.62	9.7	2080Ti	90	70.9
	LETNet [45]	2023	512×1024	No	0.95	13.6	3090	150	72.8
	FAHDNet [46]	2024	512×1024	No	0.82	9.98	2080Ti	135	70.6
	DAABNet [47]	2024	512×1024	ResNet-18	0.94	-	2080Ti	107	70.8
	LMIINet [48]	2024	512×1024	No	0.72	11.7	2080Ti	100	72.0
	HDGNet [49]	2025	512×1024	MobileNetV2	0.68	13.2	1080Ti	48	70.5
	MHMNet (ours)	-	512×1024	No	0.77	7.12	2080Ti	173	71.5

accuracy and computational efficiency.

Evaluation on Cityscapes: Quantitative comparisons with representative semantic segmentation methods on the Cityscapes test set are reported in Table 5, and the corresponding qualitative results are shown in Fig. 4. As can be observed, our MHMNet achieves 71.5% mIoU with only 0.77M parameters and 7.12G FLOPs at a resolution of 512×1024, without relying on any backbone network. Under a similar lightweight budget, MHMNet outperforms several

Table 6: Comparisons with the state-of-the-arts on the CamVid test dataset.

Methods	Year	Resolution	Backbone	Param. (M)↓	GPU	Speed (FPS)↑	mIoU (%)↑
DABNet [50]	2019	360×480	No	0.76	1080Ti	72	66.40
FDDWNet [51]	2020	360×480	No	0.80	2080Ti	79	66.90
NDNet [34]	2021	360×480	-	0.50	2080Ti	50	57.20
PDBNet [38]	2022	360×480	No	1.82	2080Ti	90	67.70
MSCFNet [40]	2022	360×480	No	1.15	Titan XP	110	69.30
AGFNet [37]	2022	360×480	No	1.12	2080Ti	100	69.40
FBSNet [44]	2023	360×480	No	0.62	2080Ti	120	68.90
FAHDNet [46]	2024	360×480	No	0.82	2080Ti	335	67.2
HDGNet [49]	2025	360×480	No	0.68	1080Ti	100	65.70
BiseNet-V1 [52]	2018	720×960	Xception	5.80	Titan XP	116	65.60
ICNet [53]	2018	720×960	PSPNet-50	26.50	Titan X	28	67.10
DFANet [28]	2019	720×960	Xception	7.80	Titan X	120	64.70
BiSeNetV2 [1]	2020	720×960	ResNet	49.00	1080Ti	60	68.70
LBN-AA [54]	2021	720×960	No	6.20	1080Ti	39	68.00
EFRNet-16 [55]	2022	720×960	EAA	1.44	2080Ti	154	68.20
SGCPNet [39]	2022	720×960	No	0.61	1080Ti	278	69.00
MLFNet [30]	2023	720×960	ResNet-34	13.03	Titan XP	57	69.00
SegTransConv [31]	2024	720×960	STDC	7.00	3090	60	68.50
MHMNet (ours)	-	360×480	No	0.77	2080Ti	292	69.50

compact CNN-based baselines, including LEDNet [32] (0.94M, 12.6G, 70.6%), LRDNet [36] (0.66M, 16.0G, 70.1%), and FAHDNet [46] (0.82M, 9.98G, 70.6%), indicating that the proposed design improves segmentation quality while remaining parameter-efficient. Compared with LMIINet [48] (0.72M, 11.7G, 72.0%), our model delivers a competitive accuracy with a substantially lower computational cost.

In terms of runtime efficiency, MHMNet runs at 173 FPS on an RTX 2080Ti GPU, showing strong real-time capability. Among the methods that also report speed on RTX 2080Ti, MHMNet is notably faster than NDNet [34] (40 FPS), SCMNet [35] (92 FPS), FBSNet [44] (90 FPS), and FAHDNet [46] (135 FPS), while maintaining favorable accuracy. Moreover, when compared to medium-sized real-time models such as BiSeNetV2 [1], which achieves 72.6% mIoU with

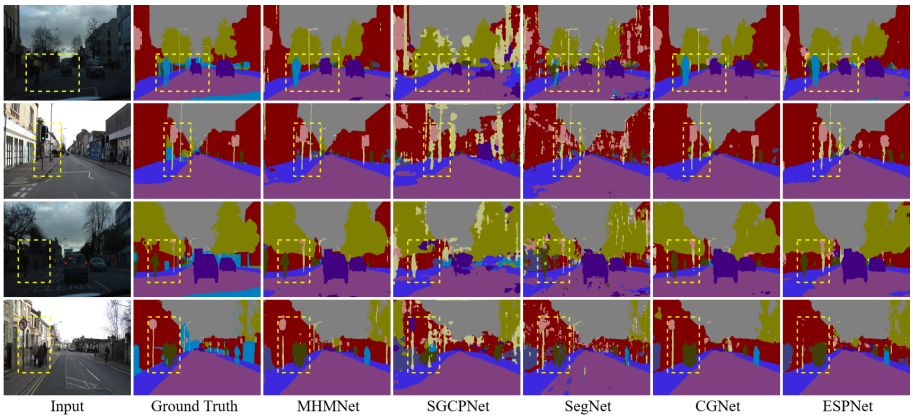

Figure 5: Visual comparisons on the CamVid dataset. From left to right are original input images, ground truths, segmentation results from our **MHMNet**, SGCPNet [39], SegNet [57], CGNet [33], and ESPNet [56].

3.40M parameters and 21.2G FLOPs, MHMNet attains comparable performance while reducing the parameter count by about 4.4× and the computational cost by about 3.0×. These comparisons further indicate that MHMNet is designed under explicit lightweight computational constraints rather than merely aiming for higher accuracy. The reported results reflect a favorable balance among model size, arithmetic cost, and throughput, suggesting that the proposed local–global integration can deliver competitive segmentation quality without exceeding tight deployment budgets. In contrast to many real-time baselines that trade speed for substantially larger parameter counts or higher FLOPs, MHMNet remains within a sub-1M parameter regime and still preserves strong real-time capability, which is particularly suitable for resource-limited and latency-sensitive settings.

Evaluation on CamVid: Table 6 summarizes the performance of MHMNet and a range of existing segmentation models on CamVid, with qualitative visualizations provided in Fig. 5. Under the lightweight setting of 360×480 input resolution, MHMNet achieves 69.50% mIoU with only 0.77M parameters and without using any backbone network. This result is competitive with other compact designs evaluated at the same resolution, such as MSCFNet [40]

(69.30%), AGFNet [37] (69.40%), and FBSNet [44] (68.90%), indicating that the proposed architecture provides strong segmentation accuracy under a strict parameter budget.

In terms of inference efficiency, MHMNet runs at 292 FPS on an RTX 2080Ti, indicating strong real-time capability. Under the same GPU setting, MHMNet is substantially faster than FDDWNet [51] (79 FPS) and PDBNet [38] (90 FPS), while also achieving higher mIoU. Although FAHDNet [46] reports a higher throughput of 335 FPS, it yields a noticeably lower accuracy (67.2%). Moreover, we include methods evaluated at a higher input resolution of 720×960, such as BiSeNet-V1 [52] and ICNet [53], which commonly rely on heavier backbone networks, as additional reference points. Despite operating at a lower resolution and with a considerably smaller model size, MHMNet remains competitive in accuracy. Overall, these results support that MHMNet provides a favorable accuracy–efficiency trade-off for real-time semantic segmentation in lightweight settings.

5. Conclusion

In this paper, we present MHMNet, a lightweight architecture tailored for real-time semantic segmentation under strict computational constraints. The framework introduces a Multi-branch Feature Aggregation Block (MFAB) in the encoder to perform efficient multi-scale feature extraction and cross-scale fusion via multi-path interactions. In the decoder, we develop a Hybrid Mamba decoding module that integrates state space modeling with local feature processing to jointly capture global contextual semantics and fine-grained details with linear computational complexity. Furthermore, a Gated Feature Selection Head (GFSH) is designed to adaptively recalibrate decoded features through channel-wise gating, enhancing discriminative responses and improving prediction stability. Extensive experiments on public benchmarks demonstrate that MHMNet achieves a favorable accuracy–efficiency trade-off, highlighting its potential for real-world applications such as autonomous driving. The current

architecture adopts a fixed local–global partitioning and directional scan aggregation scheme, whose behavior may vary when transferred to different domains or resolutions. Future work will pursue more adaptive fusion and context modeling strategies, together with deployment-aware optimization, to further improve robustness and practical efficiency.

References

[1] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, N. Sang, BiSeNet V2: Bilateral network with guided aggregation for real-time semantic segmentation, International Journal of Computer Vision 129 (2011) 3051–3068.

[2] Z. Zhang, G. Chen, X. Wang, M. Shu, Ddrnet: Fast point cloud registration network for large-scale scenes, ISPRS Journal of Photogrammetry and Remote Sensing 175 (2021) 184–198.

[3] H. Yan, C. Zhang, M. Wu, Lawin transformer: Improving semantic segmentation transformer with multi-scale representations via large window attention, arXiv preprint arXiv:2201.01615.

[4] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, P. Luo, Segformer: Simple and efficient design for semantic segmentation with transformers, Advances in Neural Information Processing Systems 34 (2021) 12077–12090.

[5] M.-H. Guo, C.-Z. Lu, Q. Hou, Z. Liu, M.-M. Cheng, S.-M. Hu, Segnext: Rethinking convolutional attention design for semantic segmentation, Advances in Neural Information Processing Systems 35 (2022) 1140–1156.

[6] J. Lee, D. Kim, J. Ponce, B. Ham, Sfnet: Learning object-aware semantic correspondence, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 2278–2287.

[7] H. Xu, C. Li, Y. Liu, G. Shao, Z. K. Hussain, A fast rooftop extraction deep learning method based on pp-liteseg for uav imaginary, in: Proceedings

of the International Conference on Artificial Intelligence and Automation Control (AIAC), IEEE, 2023, pp. 70–76.

[8] J. Xu, Z. Xiong, S. P. Bhattacharyya, Pidnet: A real-time semantic segmentation network inspired by pid controllers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 19529–19539.

[9] Y. Sun, D. Dai, Q. Zhang, Y. Wang, S. Xu, C. Lian, Msca-net: Multi-scale contextual attention network for skin lesion segmentation, Pattern Recognition 139 (2023) 109524.

[10] M.-H. Guo, Z.-N. Liu, T.-J. Mu, S.-M. Hu, Beyond self-attention: External attention using two linear layers for visual tasks, IEEE Transactions on Pattern Analysis and Machine Intelligence 45 (5) (2022) 5436–5447.

[11] J. Wang, C. Gou, Q. Wu, H. Feng, J. Han, E. Ding, J. Wang, Rtformer: Efficient design for real-time semantic segmentation with transformer, Advances in Neural Information Processing Systems 35 (2022) 7423–7436.

[12] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, X. Wang, Vision mamba: Efficient visual representation learning with bidirectional state space model, arXiv preprint arXiv:2401.09417.

[13] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, J. Jiao, Y. Liu, Vmamba: Visual state space model, Advances in Neural Information Processing Systems 37 (2024) 103031–103063.

[14] J. Ruan, J. Li, S. Xiang, Vm-unet: Vision mamba unet for medical image segmentation, arXiv preprint arXiv:2402.02491.

[15] X. Ma, X. Zhang, M.-O. Pun, Rs 3 mamba: Visual state space model for remote sensing image semantic segmentation, IEEE Geoscience and Remote Sensing Letters 21 (2024) 1–5.

[16] B. Li, X. Yang, Y. Fan, Mafmamba: A multi-scale adaptive fusion network for semantic segmentation of high-resolution remote sensing images, Sensors (Basel, Switzerland) 26 (2) (2026) 531.

[17] H. Cao, Y. Wang, J. Chen, D. Jiang, X. Zhang, Q. Tian, M. Wang, Swin-unet: Unet-like pure transformer for medical image segmentation, in: Proceedings of the European Conference on Computer Vision, Springer, 2022, pp. 205–218.

[18] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, B. Schiele, The cityscapes dataset for semantic urban scene understanding, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 3213–3223.

[19] G. J. Brostow, J. Shotton, J. Fauqueur, R. Cipolla, Segmentation and recognition using structure from motion point clouds, in: Proceedings of the European Conference on Computer Vision, Springer, 2008, pp. 44–57.

[20] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, B. Guo, Swin transformer: Hierarchical vision transformer using shifted windows, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022.

[21] M. Yang, K. Yu, C. Zhang, Z. Li, K. Yang, Denseaspp for semantic segmentation in street scenes, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 3684–3692.

[22] J. Fu, J. Liu, H. Tian, Y. Li, Y. Bao, Z. Fang, H. Lu, Dual attention network for scene segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 3146–3154.

[23] S. Zheng, J. Lu, H. Zhao, X. Zhu, Z. Luo, Y. Wang, Y. Fu, J. Feng, T. Xiang, P. H. Torr, et al., Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers, in: Proceedings of

the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 6881–6890.

[24] J. Gu, H. Kwon, D. Wang, W. Ye, M. Li, Y.-H. Chen, L. Lai, V. Chandra, D. Z. Pan, Multi-scale high-resolution vision transformer for semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 12094–12103.

[25] Z. Lai, Y. Duan, J. Dai, Z. Li, Y. Fu, H. Li, Y. Qiao, W. Wang, Denoising diffusion semantic segmentation with mask prior modeling, arXiv preprint arXiv:2306.01721.

[26] E. Romera, J. M. Alvarez, L. M. Bergasa, R. Arroyo, ERFNet: Efficient residual factorized convnet for real-time semantic segmentation, IEEE Transactions on Intelligent Transportation Systems 19 (1) (2017) 263–272.

[27] S. Mehta, M. Rastegari, L. Shapiro, H. Hajishirzi, ESPNetv2: A lightweight, power efficient, and general purpose convolutional neural network, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 9190–9200.

[28] H. Li, P. Xiong, H. Fan, J. Sun, Dfanet: Deep feature aggregation for real-time semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 9522–9531.

[29] M. Fan, S. Lai, J. Huang, X. Wei, Z. Chai, J. Luo, X. Wei, Rethinking bisenet for real-time semantic segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 9716–9725.

[30] J. Fan, F. Wang, H. Chu, X. Hu, Y. Cheng, B. Gao, MLFNet: Multi-level fusion network for real-time semantic segmentation of autonomous driving, IEEE Transactions on Intelligent Vehicles 8 (1) (2023) 756–767.

[31] J. Fan, B. Gao, Q. Ge, Y. Ran, J. Zhang, H. Chu, Segtransconv: Transformer and cnn hybrid method for real-time semantic segmentation of au-

tonomous vehicles, IEEE Transactions on Intelligent Transportation Systems 25 (2) (2024) 1586–1601.

[32] Y. Wang, Q. Zhou, J. Liu, J. Xiong, G. Gao, X. Wu, L. J. Latecki, LEDNet: A lightweight encoder-decoder network for real-time semantic segmentation, in: Proceedings of the IEEE International Conference on Image Processing (ICIP), IEEE, 2019, pp. 1860–1864.

[33] T. Wu, S. Tang, R. Zhang, J. Cao, Y. Zhang, CGNet: A light-weight context guided network for semantic segmentation, IEEE Transactions on Image Processing 30 (2020) 1169–1179.

[34] Z. Yang, H. Yu, Q. Fu, W. Sun, W. Jia, M. Sun, Z.-H. Mao, NDNet: Narrow while deep network for real-time semantic segmentation, IEEE Transactions on Intelligent Transportation Systems 22 (9) (2021) 5508–5519.

[35] T. Singha, M. Bergemann, D.-S. Pham, A. Krishna, SCMNet: Shared context mining network for real-time semantic segmentation, in: Proceedings of the Digital Image Computing: Techniques and Applications (DICTA), IEEE, 2021, pp. 1–8.

[36] M. Zhuang, X. Zhong, D. Gu, L. Feng, X. Zhong, H. Hu, LRDNet: A lightweight and efficient network with refined dual attention decorder for real-time semantic segmentation, Neurocomputing 459 (2021) 349–360.

[37] Z. Zhao, Z. Liu, C. Peng, AGFNet: Attention guided fusion network for camouflaged object detection, in: Proceedings of the CAAI International Conference on Artificial Intelligence, Springer, 2022, pp. 478–489.

[38] Y. Dai, J. Wang, J. Li, J. Li, Pdbnet: Parallel dual branch network for real-time semantic segmentation, International Journal of Control, Automation and Systems 20 (8) (2022) 2702–2711.

[39] S. Hao, Y. Zhou, Y. Guo, R. Hong, J. Cheng, M. Wang, Real-time semantic segmentation via spatial-detail guided context propagation, IEEE Transactions on Neural Networks and Learning Systems 36 (3) (2022) 4042–4053.

[40] G. Gao, G. Xu, Y. Yu, J. Xie, J. Yang, D. Yue, MSCFNet: a lightweight network with multi-scale context fusion for real-time semantic segmentation, IEEE Transactions on Intelligent Transportation Systems 23 (12) (2021) 25489–25499.

[41] Q. Lv, X. Sun, C. Chen, J. Dong, H. Zhou, Parallel complement network for real-time semantic segmentation of road scenes, IEEE Transactions on Intelligent Transportation Systems 23 (5) (2021) 4432–4444.

[42] S. Mazhar, N. Atif, M. K. Bhuyan, S. R. Ahamed, Block attention network: A lightweight deep network for real-time semantic segmentation of road scenes in resource-constrained devices, Engineering Applications of Artificial Intelligence 126 (2023) 107086.

[43] T. Singha, D.-S. Pham, A. Krishna, A real-time semantic segmentation model using iteratively shared features in multiple sub-encoders, Pattern Recognition 140 (2023) 109557.

[44] G. Gao, G. Xu, J. Li, Y. Yu, H. Lu, J. Yang, FBSNet: A fast bilateral symmetrical network for real-time semantic segmentation, IEEE Transactions on Multimedia 25 (2023) 3273–3283.

[45] N. Ta, H. Chen, X. Liu, N. Jin, Let-net: locally enhanced transformer network for medical image segmentation, Multimedia Systems 29 (6) (2023) 3847–3861.

[46] X. Hu, J. Feng, A fast attention-guided hierarchical decoding network for real-time semantic segmentation, Sensors 24 (1) (2024) 95.

[47] Q. Tang, Y. Chen, M. Zhao, S. Min, W. Jiang, Daabnet: depth-wise asymmetric attention bottleneck for real-time semantic segmentation, International Journal of Multimedia Information Retrieval 13 (1) (2024) 12.

[48] Y. Qiu, G. Xu, G. Gao, Z. Guo, Y. Yu, C.-W. Lin, Efficient semantic segmentation via lightweight multiple-information interaction network, arXiv preprint arXiv:2410.02224.

[49] Y. Zhang, X. Zhang, D. Miao, H. Yu, Real-time semantic segmentation of road scenes via hybrid dilated grouping network, International Journal of Network Dynamics and Intelligence 4 (1) (2025) 100006.

[50] G. Li, I. Yun, J. Kim, J. Kim, DABNet: Depth-wise asymmetric bottleneck for real-time semantic segmentation, arXiv preprint arXiv:1907.11357.

[51] J. Liu, Q. Zhou, Y. Qiang, B. Kang, X. Wu, B. Zheng, FDDWNet: a lightweight convolutional neural network for real-time semantic segmentation, in: Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 2373–2377.

[52] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, N. Sang, BiSeNet: Bilateral segmentation network for real-time semantic segmentation, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 325–341.

[53] H. Zhao, X. Qi, X. Shen, J. Shi, J. Jia, Icnet for real-time semantic segmentation on high-resolution images, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 405–420.

[54] G. Dong, Y. Yan, C. Shen, H. Wang, Real-time high-performance semantic image segmentation of urban street scenes, IEEE Transactions on Intelligent Transportation Systems 22 (6) (2020) 3258–3274.

[55] X. Li, F. Yang, A. Luo, Z. Jiao, H. Cheng, Z. Liu, EFRNet: Efficient feature reconstructing network for real-time scene parsing, IEEE Transactions on Multimedia 24 (2022) 2852–2865.

[56] S. Mehta, M. Rastegari, A. Caspi, L. Shapiro, H. Hajishirzi, ESPNet: Efficient spatial pyramid of dilated convolutions for semantic segmentation, in: Proceedings of the European Conference on Computer Vision (ECCV), 2018, pp. 552–568.

[57] V. Badrinarayanan, A. Kendall, R. Cipolla, SegNet: A deep convolutional encoder-decoder architecture for image segmentation, IEEE Transactions on Pattern Analysis and Machine Intelligence 39 (12) (2017) 2481–2495.